



Powering the AI Boom

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Executive Summary

Artificial intelligence is likely to become a general-purpose technology shaping production, services, and everyday life. If that happens, electricity demand will rise significantly, especially through data centers that need computing power, storage, networking, and cooling. That raises an obvious practical question: Can the U.S. system expand quickly enough to meet demand?

The short answer is yes, but not automatically. The United States enters this period with a diverse electricity system. According to the U.S. Energy Information Administration's *Electric Power Annual 2024*, natural gas remains the largest source of generation, nuclear continues to provide stable carbon-free baseload power, coal is still declining, and wind and solar have expanded quickly. Electricity demand has also grown across residential, commercial, and industrial sectors. That is a strong starting point, but not a guarantee of readiness for a sudden concentration of large-scale data center development.

The real issue is timing. Data centers can often be planned and built within two to three years. Many electricity investments cannot. Some resources, such as utility-scale solar and simple-cycle gas turbines, can be built fairly quickly. Others, including combined-cycle gas plants, geothermal projects, large transmission upgrades, hydropower, and nuclear, take much longer to plan, permit, finance, and build. This mismatch is consequential because the first constraint is typically local, not national. A region can look fine on paper and still run into serious trouble if a cluster of large facilities shows up before substations, transmission, and firm capacity are ready.

That is why the right response is not to block data center growth, but to lean into the long-run solution: capital investment and innovation. More generation, stronger transmission and distribution networks, better grid management, more efficient chips, improved cooling systems, and more resilient supply chains can all expand the system's ability to serve new demand. But those things happen only if the policy framework gives investors, utilities, technology firms, and regulators enough clarity to act before the bottlenecks become acute.

That framework should be market-oriented. Technologies should compete on cost, speed, and reliability. Large loads should face clearer rules on cost allocation, stronger incentives to contract early, and more reason to provide operational flexibility when local systems are tight. Investors should know how interconnection decisions are made, how risks are allocated, and under what conditions capital expenditures will be recoverable. Innovation should be supported by stable rules, credible price signals, access to capital, and room for competition.

Taken together:

1. AI-driven electricity demand is a real challenge, but it is fundamentally a build-and-coordinate problem, not an argument for suppressing demand.
2. The hardest issue is the mismatch between the speed of digital infrastructure development and the slower pace of energy infrastructure development.
3. The best response is a predictable, market-oriented framework that moves capital sooner, speeds innovation, and allocates risks clearly enough for investment to happen.

Bottom line for policy and workable solutions: The policy question is not whether to accommodate the AI boom, but how to get the next increment of power generation, transmission, and local grid capacity operational on a timeline consistent with demand growth and at a manageable cost.

Why This Matters Now

If AI fulfills even part of its promise, it will change how firms produce, how workers process information, and how households consume services. That should be viewed first as an economic opportunity. New technologies that improve productivity and expand useful services usually raise demand for complementary infrastructure. In this case, the most important complement is electricity.

Data centers are where that relationship becomes concrete. The digital economy may look weightless from the outside, but it rests on very physical assets: land, buildings, semiconductors, cooling systems, substations, transmission lines, and power generation. A large data center campus is not simply another office building with servers inside. In some cases, a single project can require hundreds of megawatts of electricity per year, specialized interconnection arrangements, and substantial local network upgrades. In practice, that means the AI boom reaches the power system not as a smooth national trend, but as a series of concentrated local demand surges.

This point matters because public discussion often jumps too quickly from rising demand projections to the claim that data centers are overwhelming the grid. That framing is too blunt to be useful. Demand growth is not, by itself, a sign of failure. In most sectors, new demand justifies investment in the first place. Moreover, recent gains in energy efficiency have moderated baseline demand growth, underscoring that the system can adapt when incentives are properly structured. The relevant question is whether institutions and incentives are aligned well enough for investment to respond in time.

Electricity makes that question unusually difficult. New generating assets and grid upgrades require large upfront expenditures, long lead times, and confidence that future revenues will justify capital committed today. Digital infrastructure can often move much faster. That creates the tension at the center of this essay: The technologies that need electricity often arrive faster than the infrastructure needed to serve them.

There is still no reason to treat this as an unprecedented breakdown. The U.S. power system has adjusted before to major structural increases in demand. Industrial expansion, suburban electrification, the spread of air conditioning, and earlier digital technologies all forced the system to grow beyond what existing capacity could comfortably serve. None of those transitions were frictionless, but they were resolved through investment, innovation, and institutional adaptation, not by trying to stop the underlying technologies.

The same broad logic applies here. If AI becomes economically important, electricity demand will rise. That is not the real problem. The problem is whether policy encourages timely capital formation, clearer coordination, and faster innovation. If it does, the system can adapt. If it does not, local bottlenecks, regulatory uncertainty, and slow permitting will do far more damage than demand growth itself.

Think about how this looks on the ground. A mayor may hear about a large new campus as a jobs-and-tax-base story. A utility planner may see the same project as a substation and transmission problem. A state regulator may see it as a cost-allocation and reliability question. The policy task is to connect those angles rather than let them drift into separate debates.

The Starting Point



Any serious discussion of data center electricity demand has to start with the system that already exists. The U.S. generation mix in 2024 remained dominated by natural gas, which produced 1,869,902 GWh according to the *Electric Power Annual 2024*. Nuclear generated 781,865 GWh, while coal produced 652,156 GWh. Wind reached 451,904 GWh, total solar reached 303,752 GWh, and conventional hydroelectric generation contributed 242,896 GWh. Smaller sources such as geothermal and biomass played a more modest role.

That snapshot matters because it shows both diversity and constraint. The United States is not entering the AI era with a single-fuel system or with no recent experience adding resources. At the same time, each technology does a different job. Natural gas provides dispatchable generation. Nuclear provides stable carbon-free baseload power. Wind and

solar have expanded quickly and now account for a large share of new non-fossil generation, but their contribution to reliability depends on timing, location, storage, and the rest of the system. Hydro remains important but geographically limited.

The decade-long trend tells the same story. Coal output has fallen sharply since 2014. Natural gas has grown substantially. Wind has more than doubled, and solar has risen from a relatively small base to a major contributor. Nuclear and hydro have remained comparatively stable. That history is useful because it shows that the system can change a great deal over ten years. But it also shows that the fastest-growing technologies are not always the same ones that provide the most dependable firm capacity in every hour and every place.

Demand conditions reinforce the point. Total ultimate customers rose from 147,373,702 in 2014 to 163.7 million in 2024. Total electricity sales increased from 3,764,700 GWh to 3,975,382 GWh over the same period. Residential, commercial, and industrial demand all remain large, and much of the system is already responding to population growth, housing development, and other forms of electrification. Data centers therefore do not enter an otherwise static environment. They add a large new source of load to a system that is already expanding for other reasons.

Sector-level data makes the local pressure more concrete. In 2024, residential sales reached 1,482,874 GWh, commercial sales 1,450,941 GWh, and industrial sales 1,034,584 GWh. Those numbers make clear that the system is already balancing large and economically important uses across households, offices, schools, retail, factories, warehouses, and transportation-linked activity. Data centers do not replace those uses. They sit on top of them. That is one reason local bottlenecks become politically important so quickly. When a large new customer arrives, the question is not only whether it brings investment and tax revenue, but whether the network can serve it without degrading reliability or shifting too much cost onto existing users.

When a utility territory receives several proposals for large campuses at once, local leaders start asking practical questions: Will this require a new substation? Will households end up paying for part of it? Will the project need a transmission upgrade that

takes longer than the project itself? These are exactly the issues that turn a national technology story into a local policy problem.

The implication is simple. The current generation mix and demand profile provide a useful baseline, but not one that removes the need for fast adjustment. What matters now is not simply whether the United States has enough electricity in the abstract. It is whether the right resources can be added in the right places, under the right market conditions, before concentrated AI-related demand exposes local weaknesses in the system.

Where Bottlenecks Show Up First

The most important economic fact in this discussion is that digital infrastructure and energy infrastructure move at different speeds. A data center can often be designed, permitted, and built faster than many power-sector investments. The International Energy Agency's *Energy and AI* report makes this explicit. Utility-scale solar and simple-cycle gas turbines may fit a one-to-four-year horizon. Combined-cycle gas plants, geothermal, and major grid reinforcements often require something closer to three to seven years. Large hydropower and conventional nuclear require longer still.

This mismatch has two consequences. First, the early stress from AI demand is likely to be local rather than national. The country can have adequate aggregate generation and still face serious local constraints where data center projects cluster. Second, the real question is not only how much capacity the system will eventually need, but what can plausibly be built within the next few years.

That is why a staged policy response is more realistic than a one-shot answer. In the short run, regions may rely on the resources and upgrades that can move first: solar paired with storage, quick-build gas capacity where appropriate, targeted substation work, and faster interconnection processing for projects already near completion. In the medium term, the focus shifts toward combined-cycle gas, geothermal, larger transmission work, and other investments that cannot solve next year's bottleneck but will shape conditions later in the decade. In the longer run, policy choices will help determine how much firm low-emission capacity should come from nuclear, hydro, or other durable sources. The key point is that these horizons should complement one another rather than compete for attention.

A fast-growing region may receive proposals for several large campuses that require new substations, feeder upgrades, and additional firm capacity. The transmission solution may take years, and a conventional large plant may not be online by the time the first wave of demand arrives. The near-term response may rely on faster-build resources, local network upgrades, operational coordination, and contracting arrangements that buy time until larger investments are finished. The medium-term response may look very different from the long-term one.

This is the kind of situation a state energy office or public utility commission actually faces. If a developer wants power in thirty months but the needed transmission reinforcement takes five years, the immediate questions are about what bridges the gap: a smaller generation project, phased build-out, prioritized substation work, or staggered energization. These are not glamorous questions, but they are where policy lives.

That is why sequencing matters so much. The near term is governed by what can be deployed quickly. The medium term depends on projects already moving through development queues. The long term depends on technologies that may be too slow to

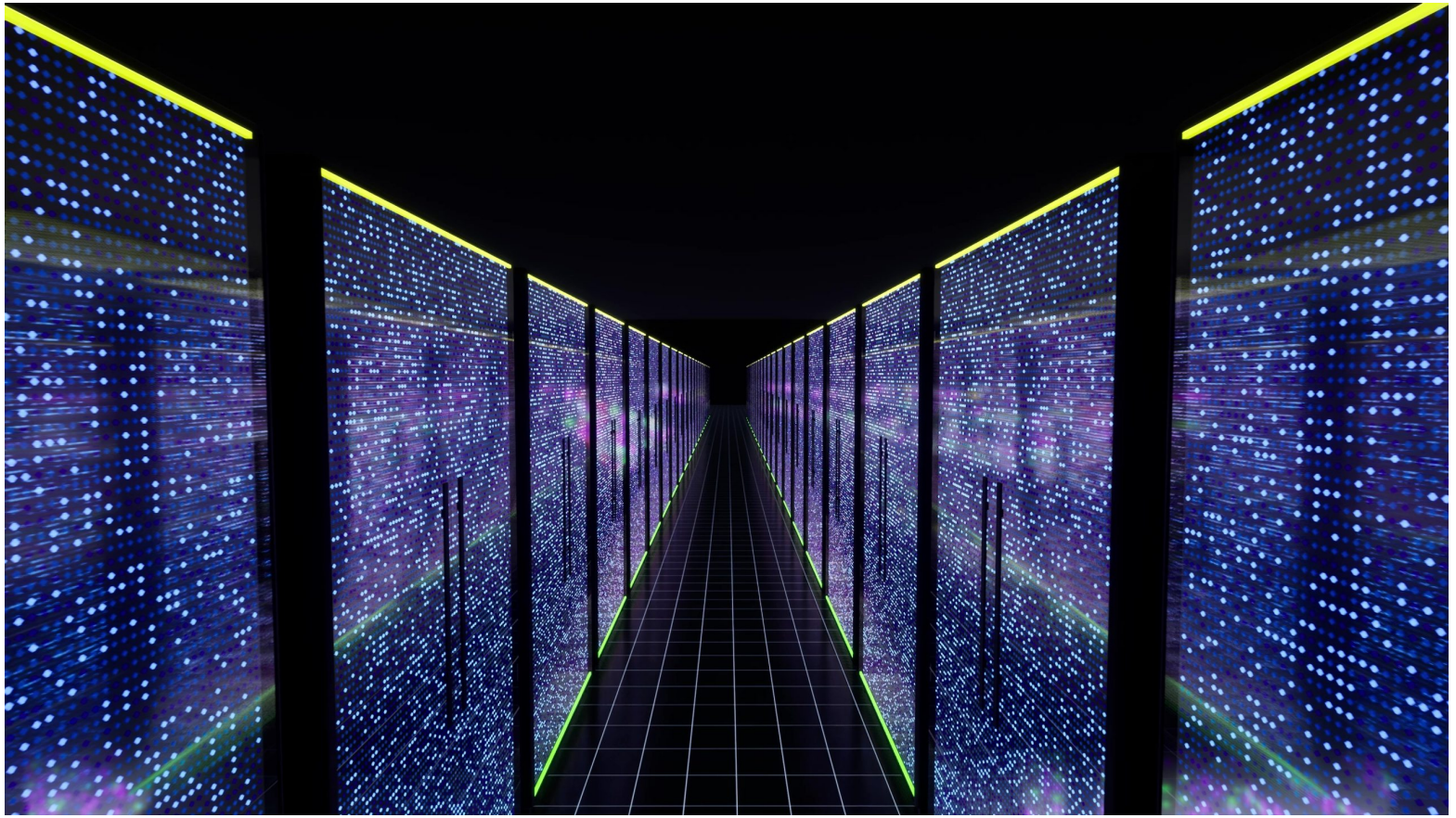
solve immediate bottlenecks but are still essential to the system that will emerge later in the decade. Policy fails when it treats those horizons as interchangeable.

The challenge is harder because the institutional environment is fragmented. Some states rely on vertically integrated utilities; others separate generation from wires and rely more heavily on organized wholesale markets. The instruments differ by region, but the economic requirement does not: Generation decisions matter only if the network can deliver power where demand is growing. A new plant does not solve a local reliability problem if transmission constraints prevent its output from reaching the load. Nor does a theoretically sound regional plan solve the problem if local permitting barriers and opposition prevent the necessary infrastructure from being built.

For that reason, the AI boom should be understood as a test of coordination as much as a test of resource adequacy. Regions that can align local permitting, interconnection, transmission planning, and large-load contracting will adjust faster than regions that cannot.



Enabling Capital Deployment



Power-sector investment is capital intensive. Projects require large upfront expenditures and produce revenues only over long time horizons. Investors must evaluate them through discounted cash flow and net present value calculations that depend on assumptions about future electricity prices, utilization rates, fuel costs, financing conditions, and the regulatory environment. AI adds another layer of uncertainty because no one knows exactly how quickly data center demand will grow or how much efficiency improvements in chips and software will offset it.

That uncertainty cuts both ways. If investors underestimate AI demand, the system may face tight conditions and high costs because too little capacity was added in time. If they overestimate it, some assets may turn out to be underutilized and politically difficult to pay for. That is why the policy objective should not be to eliminate commercial risk. It should be to reduce avoidable uncertainty so that investment decisions reflect real economic tradeoffs rather than confusion about rules.

Interest rates matter enormously in this setting. Higher borrowing costs raise the effective cost of capital and reduce the present value of long-lived revenue streams. Technologies with high upfront costs and long construction times are especially sensitive.

Regulatory uncertainty has a similar effect. If developers do not know how interconnection studies will be handled, how network upgrade costs will be allocated, whether prudently incurred expenditures will be recoverable, or how future market design will evolve, the risk premium rises, and some projects no longer look financeable.

This is where a market-oriented policy framework becomes essential. In regulated states, investors and utilities need clearer standards for prudence, cost recovery, and treatment of large-load commitments. In competitive systems, investors need organized markets and bilateral contracting arrangements that make future revenues more predictable. In both cases, transparent interconnection processes, clearer timelines, low barriers to entry, and better cost-allocation rules reduce uncertainty and speed decision-making.

Large data center developers also have a role to play. If a project requires major local upgrades, stronger load commitments, phased development milestones, and some form of cost sharing may be appropriate before existing customers are asked to bear more risk. That is not anti-investment. It is the opposite. It is a way to make large projects more bankable while reducing the chance that speculative demand projections leave households and smaller businesses paying for assets that are never fully used.

Long-term power purchase agreements, co-financing arrangements for specific upgrades, technology-neutral procurement, and clearer rules for bilateral contracting can all help. The core principle is simple: If the goal is to move capital earlier, policy needs institutions that make future revenues and future responsibilities easier to see.

There is also a straightforward governance issue in the background. In some regions, local governments want the tax base and construction activity that large data center projects bring, but they are uneasy about land use, water use, and system impacts. In others, utilities may be willing to serve new load but uncertain about how aggressively they should build ahead of demand. A better policy framework does not eliminate those tensions. It clarifies them. In a commission hearing or county meeting, that means clearer answers about who pays for triggered upgrades, what commitments the developer must make, and what happens if the project is phased, delayed, downsized, or canceled.

Why Innovation Buys Time

Investment alone will not solve the problem. Innovation matters because it changes both sides of the equation. On the supply side, it can lower construction risk, improve generation efficiency, expand storage capabilities, and improve how the grid is managed. On the demand side, it can reduce the electricity required for a given amount of computing through better chips, better cooling, and smarter software.

This point is worth stressing because the AI-electricity debate is often framed too narrowly around building more power plants. More generation will be necessary, but so will improvements in how efficiently electricity is used. A more efficient server architecture, a better cooling system, or a smarter workload management tool can reduce peak requirements and stretch existing capacity further. These are not side issues. They are part of the adjustment process.

From a policy standpoint, the issue is not abstract. A better liquid-cooling system can reduce the electricity needed to keep a facility operating safely. A more efficient chip design can lower power use per unit of computing. A workload management platform can shift certain non-urgent tasks to times when the local system is less constrained. Because data center design and operating practices are changing rapidly, these improvements may come not only from breakthrough inventions, but from continuous iteration in how facilities are built, cooled, powered, and managed. None of these developments eliminates the need for new infrastructure, but each can buy time and reduce strain.

The small modular reactor (SMR) example illustrates the broader point. According to the International Energy Agency's *The Path to a New Era for Nuclear Energy*, the first commercial SMRs are expected to begin operation around 2030. Their attraction lies not only in lower emissions, but also in the possibility of factory-based production, more standardization, and lower construction risk relative to very large custom projects. That

potential has also drawn growing interest from states seeking to create legal and regulatory frameworks for advanced nuclear deployment, including frameworks that could allow SMRs to be paired with large-scale industrial users, such as data centers.

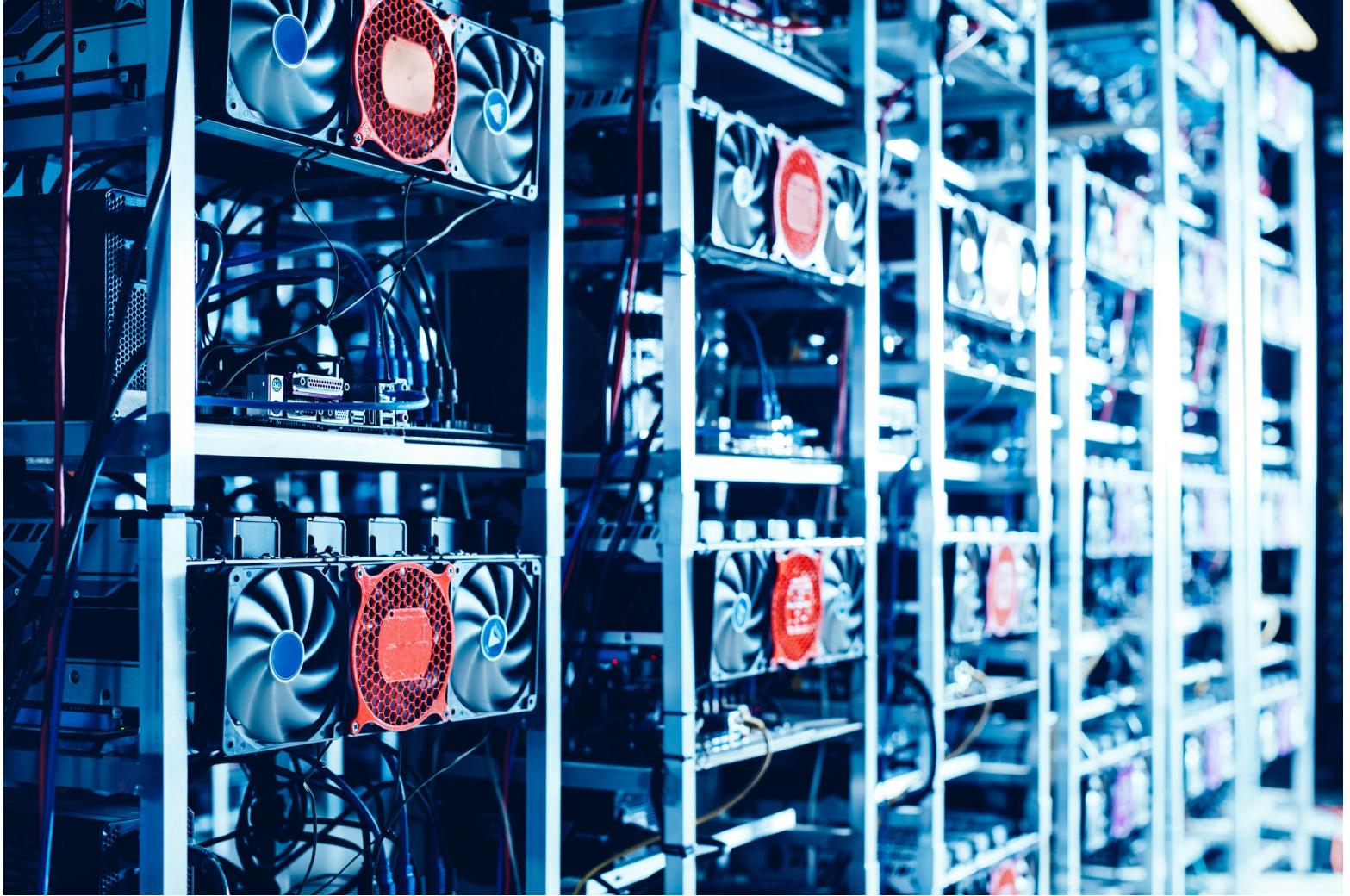
Similar logic applies outside nuclear: Modular equipment, standardized designs, and repeatable permitting processes can lower the cost and uncertainty of deployment even when the underlying technology is not new.

Innovation is also an investment problem. Research and development are costly, returns are uncertain, and commercialization takes time. Stable policy, well-defined property rights, low barriers to entry, access to capital, and competition all affect whether promising ideas are scaled or abandoned. If the policy environment is volatile or opaque, firms will underinvest not only in new infrastructure but also in the technologies that make that infrastructure cheaper and more effective. Innovation depends not only on technical possibility, but on the market conditions that tell firms which problems are worth solving.

That is part of why credible price signals matter, even if they are not the whole story. Investors and innovators respond to scarcity, expected demand growth, and the value of flexibility. Competition matters too. Open entry and room for experimentation increase the likelihood that better solutions are discovered and adopted. In that sense, innovation should not be treated as an afterthought to capital formation. It is part of the same process.

The timing argument applies to demand-side innovation too. Advances in chip design, server architecture, cooling technology, and energy management systems can materially reduce the electricity needed for a given amount of computing. Even if AI demand keeps rising, its energy intensity can be moderated. Those improvements are valuable precisely because they affect how quickly local systems become constrained. They do not replace new generation and new wires, but they can ease the transition while larger investments are being built.

What to Watch Beyond the Grid



The AI-energy transition is not purely domestic. Data centers depend on semiconductors, transformers, switchgear, cooling equipment, and other inputs that move through global supply chains, so competition for these components is likely to intensify as more countries work to expand digital and energy capacity at once.

Even well-designed domestic plans may be delayed by transformer shortages, chip export restrictions, and backlogged electrical equipment. A utility may be ready for a substation upgrade while waiting months for a transformer; a developer may have land and financing but still lack key equipment. The point is not that we should aim at self-

sufficiency. It is that resilient sourcing, stable commercial relations, supplier diversification, and more flexible procurement can reduce risk without turning energy and AI policy into protectionism.

In the end, AI-driven electricity demand is a manageable challenge rather than a reason for alarm. The central issue is not whether a valuable new source of demand should exist, but whether energy supply, grid capacity, and supporting technologies can expand quickly enough to meet it. That expansion will not happen automatically. It will require investment, innovation, clearer rules, credible price signals, and institutions capable of moving at the speed of the technologies they are being asked to support.



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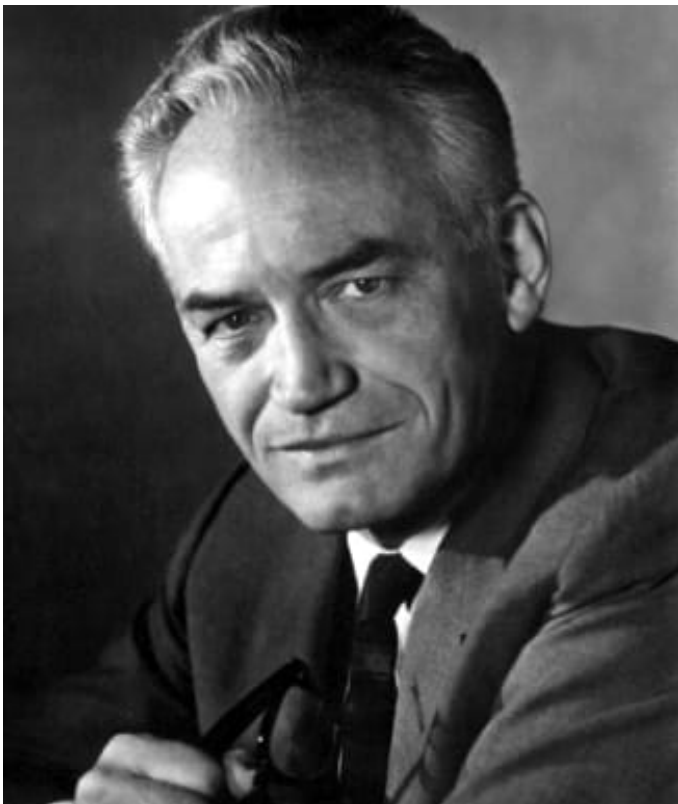
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